

Task1 - Image Inpainting

1 Overview

The inpainting model is a Generative Adversarial Network (GAN), structured after the Pix2Pix image-to-image translation framework (Isola et al., 2017). It consists of a U-Net generator that reconstructs a full 64×64 face image from a masked input, and a PatchGAN discriminator that judges the realism of the completion conditioned on that same masked input. Random rectangular masks covering 10–30% of the image area are generated on-the-fly by the dataloader.

2 Masking Strategy

A custom MaskedCelebADataset wraps the base dataset and generates a random rectangular mask on the fly for every sample. Mask height and width are drawn uniformly at random and accepted via rejection sampling only if the resulting area covers between 10% and 30% of the image. The mask is returned as a separate binary tensor (1 = masked, 0 = visible) alongside the masked image and the original image. Figure 1 shows example triplets (original / mask / masked input) produced by the data loader; mask position and aspect ratio vary from sample to sample, confirming that the on-the-fly masking is working as intended.



Figure 1: Example outputs of the data loader

3 Model Architecture

3.1 Generator

The generator is a compact U-Net. It takes a 4-channel input (the 3-channel masked image concatenated with the 1-channel mask) and encodes it through four strided convolutions ($64 \rightarrow 128 \rightarrow 256 \rightarrow 512$ channels, downsampling 64×64 to 4×4). A symmetric decoder of four transposed convolutions upsamples back to 64×64 , with skip connections concatenating each decoder stage to the matching encoder feature map. The skip connections let the network copy fine spatial detail (edges, texture, color) directly from the visible, unmasked pixels instead of

forcing all information through the 4×4 bottleneck, which is important for inpainting because most of the image is already known and only needs to be passed through, not regenerated.

3.2 Discriminator

The discriminator follows the PatchGAN design used in Pix2Pix. It receives the concatenation of the conditioning signal (masked image + mask) and a candidate image (real or generated), and outputs a 7×7 grid of validity scores rather than a single real/fake decision. Each output unit corresponds to a receptive field covering a local patch of the input, so the discriminator penalises local texture around the inpainted region specifically, which is more informative for an inpainting task than a single global real/fake judgement.

4 Loss Function

4.1 Discriminator loss

The discriminator is trained to score real (masked-image, mask, ground-truth) triples as "real" and (masked-image, mask, generated-image) triples as "fake," using a standard binary cross-entropy comparison averaged over its 7×7 grid of patch predictions, so that it keeps learning to tell the generator's completions apart from genuine photos while the generator simultaneously tries to fool it.

4.2 Generator loss

Three loss terms are combined to train the generator:

$$\mathcal{L} = \mathcal{L}_{\text{GAN}} + 10 \cdot \mathcal{L}_{1,\text{global}} + 100 \cdot \mathcal{L}_{1,\text{masked}} \quad (1)$$

4.2.1 Adversarial loss (BCEWithLogitsLoss)

The discriminator is trained to output high logits for real (masked-image, mask, ground-truth) triples and low logits for (masked-image, mask, generated) triples; the generator is trained to push the discriminator's response to its own outputs toward the "real" label.

4.2.2 Global L1 loss

Computed as the mean absolute error between the generated and ground-truth image over all 64×64 pixels and all 3 channels, with no masking applied. Its role is to enforce fidelity in the region that was never masked. Skip connections make it easy for the network to pass the unmasked context through unchanged, but nothing in the architecture forces that behaviour by itself.

4.2.3 Masked L1 loss

Computed as the sum of absolute pixel errors inside the masked region only, divided by the number of masked pixels in the batch (plus a small epsilon for numerical safety). This is the core task-specific loss: because mask area varies from sample to sample (10–30% of the image), normalising by the mask's own pixel count — rather than by the full 4096-pixel image — yields a fair, comparable per-masked-pixel error regardless of hole size; without this normalisation, larger holes would trivially report larger loss values and receive disproportionately diluted gradients. This term dominates the total loss numerically (weight 100 vs. 10 vs. 1), reflecting that accurate reconstruction inside the hole is the primary objective of the task, with realism (adversarial) and context fidelity (global L1) acting as secondary, supporting objectives.

5 Results

5.1 Quantitative Results

The model was retrained for 5 epochs (636 batches/epoch) using a proper three-way split: 162,770 training images, a 19,867-image held-out validation set checked at the end of every epoch, and the official CelebA test partition (19,962 images) reserved for a final, fully independent check.

Epoch	Val. masked L1	Val. masked PSNR
1	0.5911	11.99 dB
2	0.5361	12.74 dB
3	0.5285	12.93 dB
4	0.5078	13.09 dB
5	0.5164	12.85 dB

Table 1: Validation split metrics

Validation masked L1 improves monotonically and substantially for the first four epochs ($0.591 \rightarrow 0.536 \rightarrow 0.529 \rightarrow 0.508$), then increases again at epoch 5 (0.516), the model’s held-out performance briefly regressed rather than continuing to improve. This coincides with the discriminator loss spending long stretches at very low values in epoch 5 (repeatedly 0.03–0.08 across consecutive logged batches), consistent with the discriminator beginning to dominate the generator around this point, which plausibly explains the regression.

Metric	Value (masked region)
Mean L1 (range $[-1, 1]$)	0.5088
Mean PSNR	13.08 dB

Table 2: Quantitative results on the held-out CelebA test split.

Running the epoch-4 checkpoint (our best out of 5 epochs) on the fully held-out CelebA test split (19,962 images, never seen during training or validation) gives a mean masked-region L1 of 0.5088 and mean masked-region PSNR of 13.08 dB, nearly identical to the validation numbers at that same checkpoint (0.5078 / 13.09 dB). This close match is direct evidence that the model generalises rather than overfitting to the training or validation data.

5.2 Qualitative Results

Qualitatively, the composited outputs preserve the visible pixels exactly (by construction) and produce a smoothed, colour-consistent fill for the masked region, which is more successful when the mask covers a relatively uniform area (skin, hair, background) where low-frequency colour/texture is the main requirement.

Success cases: Column 2 - figure 2: The mask here covers a genuinely difficult, high-frequency part of the face — the nose, mouth, and part of the hair — rather than a smooth area like skin or hair. Despite that, the composited output reconstructs this region acceptably. The model recovers plausible mouth shape and hair texture that blends naturally with the surrounding skin tone and lighting, without an obvious seam at the mask boundary.

column 3 - figure 2: In this example the mask covers only the neck which is a smooth, low-frequency area with no fine structure to recover. The composited result achieves a low masked-region and is visually indistinguishable from the original.

Failure cases: Column 4 - figure 2: the raw generator output and composited result are visibly blurry and muddy, the reconstructed eye/nose region is asymmetric and lacks defined

structure, and the fill doesn't recover coherent facial geometry.

column 1 - figure 2: Here the model produces a locally coherent, non-blurry result, but gets the actual semantic content of the eyes wrong. It highlights that masked-region L1 alone can understate failures that are concentrated in a small but perceptually critical sub-area (the eyes carry most of a face's expression), once a mask removes the eyes entirely, the network has no shortcut through the skip connections for that specific content and must guess a plausible-looking eye region from context alone — and here it guesses wrong.



Figure 2: Qualitative grid on held-out test images

6 Effect of Architecture and Loss Design

The U-Net's skip connections explain why visible content is always reconstructed perfectly, but they also mean the network's capacity to hallucinate missing structure is concentrated entirely in the bottleneck and decoder path. When the masked region is small or low-frequency, the skip connections carry enough surrounding context that the bottleneck barely needs to invent anything. When the masked region is large and covers high-frequency features (eyes, nose), the bottleneck has to reconstruct structure with very little local context to anchor it, which is exactly where failures concentrate.

The PatchGAN discriminator's local receptive field should, in principle, penalise blurriness and seams specifically, but its effectiveness depends on it staying in a competitive balance with the generator. Once the discriminator's loss collapses toward zero, it stops supplying a useful gradient, and the generator effectively trains under L1 supervision alone for the remainder of training. This is consistent with why even the success case still looks smoothed rather than photorealistic. The model is doing the best it can under an effectively L1-dominated objective, and it succeeds when that's sufficient (low/medium frequency content) and fails when it isn't (high-frequency structure under poor lighting).